

## Education

**University of Utah**  
*B.S. Computer Science*

**Salt Lake City, UT**  
*August 2020 – May 2025*

**Relevant Coursework:** Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

## Skills

**Languages:** Python, TypeScript, C++ , SQL, Java

**Software:** Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

## Experience

**L3Harris Technologies**  
*Associate Software Engineer*

**Dallas, TX**  
*June 2025 – Present*

- Developed real-time signal processing algorithms in Python and C++ on embedded systems, applying DSP techniques for defense sensor data streams.
- Built microservices for high-frequency data ingestion and storage, leveraging Podman and serverless infrastructure on AWS.
- Automated CI/CD pipelines with static code analysis and integration testing.

**Doxy.me**  
*Software Engineer*

**Charleston, SC**  
*June 2023 – May 2025*

- Improved clinical inference quality by building and fine-tuning transformer-based models (PyTorch, Llama) in Python and TypeScript, integrating outputs with WebRTC video pipelines and deploying end-to-end inference infrastructure on AWS SageMaker and Fargate.
- Strengthened system correctness by authoring comprehensive Python test suites covering data ingestion, model execution, and API response validation, catching regressions before they reached production.
- Reduced deployment errors by owning CI/CD configuration and infrastructure automation on AWS, enabling reproducible builds and reliable delivery of backend services across multiple environments.

**University of Utah**  
*Undergraduate Research Assistant – FuTURES Lab*

**Salt Lake City, UT**  
*May 2024 – Jun 2025*

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

## Projects

**Configuration Fuzzer:** Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

**Telehealth Mental Health Diagnosis Tool (Doxy.me):** Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

**University Course Planning/Review Platform (Class Best):** Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.